Experiment 3 Report
BCI 初應用與資料蒐集

Group 5

B12901057 楊程亘
B12901059 陳裕穎
B12901069 段奕鳴

1 Introduction

A Brain-Computer Interface (BCI) is a technology that enables direct communication between the brain and an external device. Electroencephalography (EEG) is one of the most widely used signal sources for BCIs due to its non-invasive nature, and a large part of BCI research is about classifying short EEG segments into mental states [1]. In this experiment, EEG signals are acquired with a BrainLink headset, which provides a single dry electrode placed at Fp1 over the prefrontal cortex. The objective is to apply machine learning techniques to these single-channel signals in order to distinguish between three mental states: "Relax" (eyes closed), "Focus" (watching a static dot and pressing a key when it dims) and "Stroop" (choosing the ink colour of incongruent colour words with one of four keys, without speaking [2]).

The experiment is organized into two parts. In Part 1, each group records its own dataset. In Part 2, these recordings are used to train and evaluate classifiers.

Our task is to design an effective preprocessing and feature extraction pipeline, to implement and compare at least four classifiers (MLP, Support Vector Machine, Random Forest, and one self-selected model), and to apply post-processing strategies, with the goal of raising the overall mean accuracy of the best model to 75% or higher. All models are assessed under Leave-One-Subject-Out (LOSO) cross-validation.

2 Methodology for Model Optimization

This section describes the self-recorded dataset, the data processing pipeline, the hyperparameter tuning of the four classifiers, and the performance evaluation method used to optimize the models on our 3-member group dataset.

2.1 Dataset and Environment

- **Data Source**: `bci_dataset_115-1`, containing self-recorded data from our 3 group members (30 recordings per subject, 90 in total). B12901057 and B12901069 recorded the protocol more than once. The spectra of the first recordings did not match the examples in the course material or the typical EEG spectrum, and the first session was likely disturbed by

a crowded, noisy room and a tired subject, so we recorded again. The dataset contains one complete session of each member.

- **Signal Type**: Single-channel (Fp1) EEG time-domain signal recorded via BrainLink with a dry electrode.
- **Sampling Rate**: 512 Hz; each recording covers one 20 s block (10240 samples).
- **Labels**: Each subject has ten rounds of recordings for three states (`_1_` = Relax, `_2_` = Focus, `_3_` = Stroop).
- **Environment**: Python 3.11 managed via `uv`, with scikit-learn [3], NumPy and SciPy.

2.2 Data Preprocessing and Feature Engineering

All processing is implemented inside the marked student blocks of `main.py` (Part 1 preprocessing, Part 2 feature engineering, the self-selected model block) and in the `Config` class. The classifier wrapper, the LOSO loop and the reporting code are unchanged, and all four classifiers share exactly the same feature matrix.

- **Data Segmentation:**
  - `SEGMENT_LENGTH`: **8 s** (4096 samples). Because the band powers are computed on 1 s windows inside each segment (see Feature Engineering below), the frequency resolution is fixed at 1 Hz (512-point Welch window at 512 Hz) and does not depend on the segment length. This already gives 4 to 10 frequency bins per band, for example 4 for theta and 5 for alpha. A single spectrum over the whole 8 s segment would give 0.125 Hz, which is finer than the band features need and averages short bursts away. The segment length therefore trades the number of 1 s windows per segment (29 at 8 s, which makes the distribution statistics more stable) against the number of training samples. In a sweep over 3–10 s (Table 15 in Appendix B) the mean accuracy rose from 0.72 at 3 s to about 0.80–0.85 at 8–10 s for the best model; we chose 8 s because it still provides 210 samples per subject.
  - `OVERLAP_RATIO`: **0.75** (2 s hop). Each 20 s recording yields 7 segments, i.e. 70 segments per class and 210 per subject (630 in total). Overlap increases the number of training samples without discarding data, but neighbouring segments are correlated, so they do not add fully independent information.
- **Preprocessing Pipeline:** applied to every 8 s segment, in this order:
  1. **DC removal**: subtract the segment mean to remove the electrode offset.
  2. **60 Hz notch filter** (IIR notch, $Q = 30$) to suppress mains interference (60 Hz in Taiwan).
  3. **Band-pass filter 1–45 Hz** (4th-order Butterworth, applied forward and backward so that the phase is zero) to remove slow drift and to keep the delta to gamma range.

- **Feature Engineering:** a single Welch spectrum over 8 s averages away short bursts (for example alpha bursts or a keypress artifact). Therefore, inside each segment we slide a 1 s window with a 0.25 s hop, compute 8 features per window, and describe the resulting series by its distribution (mean, standard deviation, 90th percentile and maximum). This gives $8 \times 4 = 32$ features per segment.
  - **Frequency-Domain Features**: for each window a Welch PSD [4] (one 512-point Hann window covering the 1 s window) is computed, and we take the relative power of theta (4–8 Hz), alpha (8–13 Hz), low beta (13–20 Hz) and high beta (20–30 Hz) with respect to the total 1–45 Hz power, plus $\log_{10}(\theta/\beta)$ and $\log_{10}(\alpha/\beta)$ with β = low + high beta. The bands were chosen from the usual physiological picture: alpha rises when the eyes are closed and the subject is at rest [5], beta grows with engagement and the ratios between beta and the slower bands have long been used as an engagement index [6], and frontal theta increases with cognitive control, as in the Stroop conflict [7]. Ratios and relative powers are used instead of absolute powers so that the features do not depend on the overall gain of the dry electrode.
  - **Time-Domain Features**: $\log_{10}$ of the window standard deviation (overall amplitude) and kurtosis (spikiness). They contribute information that the normalized spectrum cannot give. In the Stroop task the subject reads words and answers with the keyboard, which may add eye movement and muscle activity at Fp1; in our data Stroop indeed has the largest amplitude and kurtosis of the three classes (Section 3). The ablation in Table 4 shows that they add about 10 percentage points before subject-wise normalization.
  - **Feature Scaling**: two stages. (1) *Subject-wise z-score* inside Part 2: the 210 feature vectors of each subject are standardized with that subject's own mean and standard deviation (no labels are used), which removes between-subject differences in electrode contact, gain and baseline amplitude. Because `extract_features` only receives segments and no subject ID, the segment-to-feature table is built once per configuration from all recordings, and each subject is then normalized. (2) The framework's `StandardScaler` is fitted on the training subjects only.
  - **Feature Selection:**
    - `FEATURE_SELECTION`: **False**.
    - `N_FEATURES_SELECT`: not used (left at 20). With only 32 features, enabling feature selection with $k = 8$, 15 or 20 lowered the best accuracy from 0.825 to 0.783, 0.810 and 0.811, respectively (Table 16 in Appendix B), so all 32 features are kept.

2.3 Classifiers and Hyperparameter Experiments

This experiment requires implementing and comparing at least four classifiers: `MLPClassifier`, `SVC`, `RandomForestClassifier`, and one self-selected model. As the fourth classifier we chose **shrinkage Linear Discriminant Analysis** (LDA, `solver='lsqr'`, `shrinkage='auto'`). It is a linear model with few parameters, and the automatic Ledoit–Wolf shrinkage [8] regularizes the covariance estimate, which suits a small training set (420 samples, 32 features per LOSO fold). LDA is also a standard and often strong classifier in EEG BCIs [1]. Besides LDA, we compared QDA, logistic regression, k-NN, naive Bayes, ridge and linear SVM, gradient boosting and extra trees under the same protocol (Table 14); none was clearly better than LDA.

The hyperparameters of the three given classifiers were determined by **parameter experiments** rather than by hand. For each classifier a script in `exp_testing/` first varies one parameter at a time around the final setting, starting from the framework's initial values (`main_original.py`), and then evaluates a joint grid of the parameters that mattered most: hidden layers × learning rate × alpha for the MLP (125 settings), kernel × C for the SVM (48 settings) and number of trees × maximum depth for the Random Forest (28 settings). Every setting is scored by the mean LOSO accuracy, and the setting with the highest score is used in `main.py`. Because the MLP and the Random Forest depend on the random seed, their joint grids average 3 and 5 seeds, respectively, so that the choice is not decided by a lucky seed. All results are listed in Tables 8–13 in Appendix B. The experiments show that the three models are not very sensitive to most parameters: 39 of the 125 MLP settings are within one percentage point of the best one (0.799), the Random Forest stays within 0.74–0.76, and the number of MLP training iterations has no effect because early stopping ends the training first. The SVM is the exception: a large `C` over-fits (0.738 at $C = 10$), and C between 0.001 and 0.05 gives the best accuracy (0.811). In this range the accuracy no longer changes because the margin is so soft that every training segment becomes a support vector at the bound (at $C = 0.01$, all 420 training segments in each of the three folds), so the SVM behaves much like a kernel-weighted comparison with the class means. The settings were selected on the same three subjects that are used for the evaluation, so the resulting accuracies are slightly optimistic. The final settings are listed below.

2.4 Post-processing

Not used in the submitted code. The `POSTPROCESSING` block of `predict` is left empty, so every 8 s segment is classified independently by the model's output. We nevertheless evaluated several candidate methods offline, using the same trained models and the same LOSO folds, and did not adopt them for the reasons below. All methods only change the predicted labels of the held-out member and leave the classifiers untouched. Table 2 lists the mean LOSO accuracy.

- **Moving average over neighbouring rows:** not valid. `predict` receives the test matrix ordered by class, so a window that runs across the boundary between two recordings mixes in segments that are known to share the label. When we shuffled the rows of the test member, the accuracy of LDA with a 5-row moving average dropped from 0.825 to 0.451, so the gain comes from the order of the labels and not from the signal.
- **Voting within one recording:** valid, but not used in the submitted code. The 7 segments of one file always come from the same 20 s block and therefore share one label, so averaging their probabilities uses no label information. We checked this by shuffling the order of whole recordings inside the test member: the accuracies were identical (LDA 0.867, SVM 0.856). Per subject, LDA reached 0.767, 0.967 and 0.867. The decision is then made once per 20 s block instead of once per 8 s segment, which changes what the reported accuracy means, and `predict` only receives the feature matrix without a recording index. We therefore report it here as the expected gain of a block-level decision and keep segment-level predictions in `main.py`.
- **kNN smoothing, label propagation and equal class share:** order-independent by construction (for kNN smoothing we verified that shuffling the rows gives identical results),

Table 1: Final Hyperparameter Settings of the Four Classifiers (selected by the parameter experiments)

Model	**Parameter**	**Final Setting**	**Justification from the parameter experiments**
MLP	`HIDDEN_LAYER_SIZES`	(256, 128, 64)	Best of the joint grid (0.799, mean of 3 seeds). One layer gave 0.741–0.794 and two to four layers 0.757–0.798 in the one-at-a-time sweep, so the depth is not critical.
	`LEARNING_RATE_INIT`	0.0005	Part of the best joint setting; the one-at-a-time sweep from 0.0001 to 0.01 gave 0.775–0.798 (framework initial value: 0.002).
	`ALPHA` (L2)	0.05	Part of the best joint setting; the L2 strength from 0 to 0.5 changed the accuracy by less than 0.01.
	`MAX_ITER`	150	10 to 1000 iterations gave the same accuracy (early stopping), so the initial value is kept.
SVM	`kernel`	rbf	Best kernel: 0.811, versus 0.800 (linear), 0.783 (sigmoid) and 0.510 (polynomial).
	`C`	0.01	Accuracy is flat (0.811) for C from 0.001 to 0.05, so we took the middle of this range; it falls to 0.781 at the initial value $C = 1$ and 0.738 at $C = 10$.
	`gamma`	`'scale'` (default)	Not tuned: the default adapts to the standardized features, and γ between 0.003 and 0.03 gave 0.792–0.813.
RF	`n_estimators`	500	Best of the joint grid (0.760, mean of 5 seeds); the accuracy rises slowly with the number of trees (0.737 at 50 and 0.749 at 500 for unlimited depth).
	`max_depth`	5	Shallow trees (depth 3–5) gave 0.749–0.760 for 50–500 trees, better than unlimited depth (0.737–0.749), presumably because they over-fit less on 420 samples.
Shrinkage LDA	`solver`, `shrinkage`	`'lsqr'`, `'auto'`	Ledoit–Wolf shrinkage chooses the regularization from the data. A hand-picked value of 0.05 gave 0.827 versus 0.825, a difference that is within the noise, so we kept the automatic choice.

and they raised the accuracy of some models by 1–3 points. However, the gain is small and inconsistent across models and folds (for example, label propagation lowered LDA on B12901069 from 0.819 to 0.805), they use the unlabeled test data of the held-out member (as our z-score does), and only three folds are available to judge them. Since the best model already exceeds 80% without them, we kept the simpler, unprocessed prediction.

Table 2: Candidate post-processing methods (mean LOSO accuracy, final models)

Method	**MLP**	**SVM**	**RF**	**LDA**
None (`predict`, submitted)	0.795	0.811	0.751	0.825
Arg-max of the class probabilities (reference for the rows below)	0.795	0.803	0.751	0.825
Equal class share (re-weight the probabilities so each class gets about 1/3)	0.798	0.781	0.754	0.813
kNN smoothing ($k = 10$, average the probabilities of the nearest test segments in feature space)	0.802	0.817	0.770	0.833
Label propagation on the kNN graph ($k = 10$)	0.811	0.817	0.763	0.832
Label propagation + equal class share	0.824	0.819	0.781	0.856
Average of the probabilities over the 7 segments of each 20 s recording	0.833	0.856	0.789	0.867
Moving average over 9 neighbouring rows (*not valid*, see text)	0.837	0.843	0.784	0.875

- **Probability thresholds and clustering of the test segments:** not used. Rejecting low-confidence segments would leave them without a prediction, while the evaluation needs a label for every segment; and clustering the test segments (cluster-wise voting, a Gaussian mixture initialized from the predictions) was unstable: the Gaussian mixture raised the MLP, SVM and RF by about 2–7 points but lowered LDA to 0.790, with B12901069 falling to 0.595.

2.5 Performance Evaluation Method

All four classifiers are evaluated with **LOSO Cross-Validation** to assess their generalization ability across different users. In this method, the data of one group member is used as the test set, while the data from the remaining 2 group members are used for training. This process is repeated 3 times, with each member serving as the test set once, and the reported result is the mean accuracy with its standard deviation across the three folds. In addition to overall accuracy, the per-class precision and recall and the 3×3 confusion matrix of the best model are reported in Section 3.

3 Results and Analysis

3.1 Comparison of the Four Classifiers

All four classifiers were trained on the same feature set and evaluated with LOSO cross-validation over the 3 group members. Table 3 summarizes the overall mean accuracy and its standard deviation across the three folds.

Table 3: LOSO cross-validation accuracy of the four classifiers

Model	**Mean Accuracy**	**Std.**	**Remarks**
MLP	0.795	± 0.099	Per subject (B12901057 / 059 / 069): 0.676 / 0.919 / 0.790.
SVM	0.811	± 0.097	Second best, with $C = 0.01$; per subject: 0.729 / 0.948 / 0.757.
RF	0.751	± 0.121	Lowest of the four; per subject: 0.624 / 0.914 / 0.714.
Shrinkage LDA (self-selected)	**0.825**	± 0.101	Best model; per subject: 0.705 / 0.952 / 0.819.

Shrinkage LDA performed best with a mean accuracy of 0.825, which is 1.4, 3.0 and 7.4 percentage points higher than the SVM, MLP and RF, respectively. The hyperparameters of the three given classifiers were selected by the parameter experiments of Section 2, so these numbers are the best settings found, not the framework's initial values; the MLP and the Random Forest average 0.793 ± 0.009 and 0.758 ± 0.006 over ten random seeds (Appendix B), so their numbers vary by about one percentage point with the seed. All four classifiers reached the 75% target, and the best two (LDA and SVM) also exceed 80%. The standard deviation is the (population) standard deviation of the three fold accuracies, i.e. it reflects the difference between the three members rather than the randomness of a single run.

3.2 Effect of Preprocessing and Feature Engineering

To quantify how much of the performance comes from the processing pipeline rather than from the classifier itself, we compare a reference configuration against our final pipeline. The reference configuration feeds the raw 8 s time-domain samples (4096 values per segment) directly to the four classifiers, with no filtering and no normalization. We then add exactly one step at a time: band-pass filtering of the raw samples, the window-based band-power features, the time-domain features, and finally the subject-wise z-score, which stays off until the last step. All other settings (segmentation, hyperparameters, LOSO folds) are identical in every row. For every row the table reports the accuracy of the best of the four models, which is a different model in different rows (SVM, LDA, MLP, LDA), so the accuracy of every model is given in Appendix B. The script `exp_testing/ablation.py` in the repository reproduces this table.

Table 4: Ablation study of the processing pipeline (accuracy of the best model)

Configuration	**Mean Accuracy**	**Observation**
Raw time-domain samples	0.500 (SVM)	Reference; 4096-dimensional input, chance level is 0.333.
+ Band-pass filtering (1–45 Hz, 60 Hz notch)	0.537 (SVM)	+3.7 points; the raw samples remain high-dimensional and phase-dependent.
+ Window band-power features	0.571 (LDA)	+3.4 points; compact spectral features are far better than raw samples.
+ Time-domain features (std, kurtosis)	0.700 (MLP)	+12.9 points; amplitude and spikiness separate Stroop from the others.
+ Subject-wise z-score (final pipeline)	**0.825** (LDA)	+12.5 points; as large as the time-domain features.
Extra: band power + z-score	0.803 (LDA)	Not a cumulative step; shows that the z-score alone lifts the band-power features from 0.571 to 0.803, and the time-domain features add the last 2.2 points.

3.3 Detailed Results of the Best Model

Figure 1 shows the results of our final run.

The terminal output summary of the same run (`results_summary.txt`) is:

```
Summary of Classification Results (LOSO):
 - MLP         : 0.795 +/- 0.099
 - SVM         : 0.811 +/- 0.097
 - RF          : 0.751 +/- 0.121
 - SelfSelected: 0.825 +/- 0.101

Best Performing Model: SelfSelected with Accuracy 0.825
  -> reached the 75% target

Accuracy per held-out subject:
Subject                   MLP           SVM            RF  SelfSelected
b12901057               0.676         0.729         0.624         0.705
b12901059               0.919         0.948         0.914         0.952
b12901069               0.790         0.757         0.714         0.819

Detailed Class Report for Best Model [SelfSelected]:
[Relax Class]
  - Accuracy (Recall): 0.781 (164/210)
  - Precision: 0.820 (164/200)
  - F1-score: 0.800

[Focus Class]
```

```
  - Accuracy (Recall): 0.829 (174/210)
  - Precision: 0.777 (174/224)
  - F1-score: 0.802

[Stroop Class]
  - Accuracy (Recall): 0.867 (182/210)
  - Precision: 0.883 (182/206)
  - F1-score: 0.875

Overall Confusion Matrix [SelfSelected] (rows = true, cols = predicted):
               Relax     Focus    Stroop
Relax            164        34        12
Focus             24       174        12
Stroop            12        16       182
```

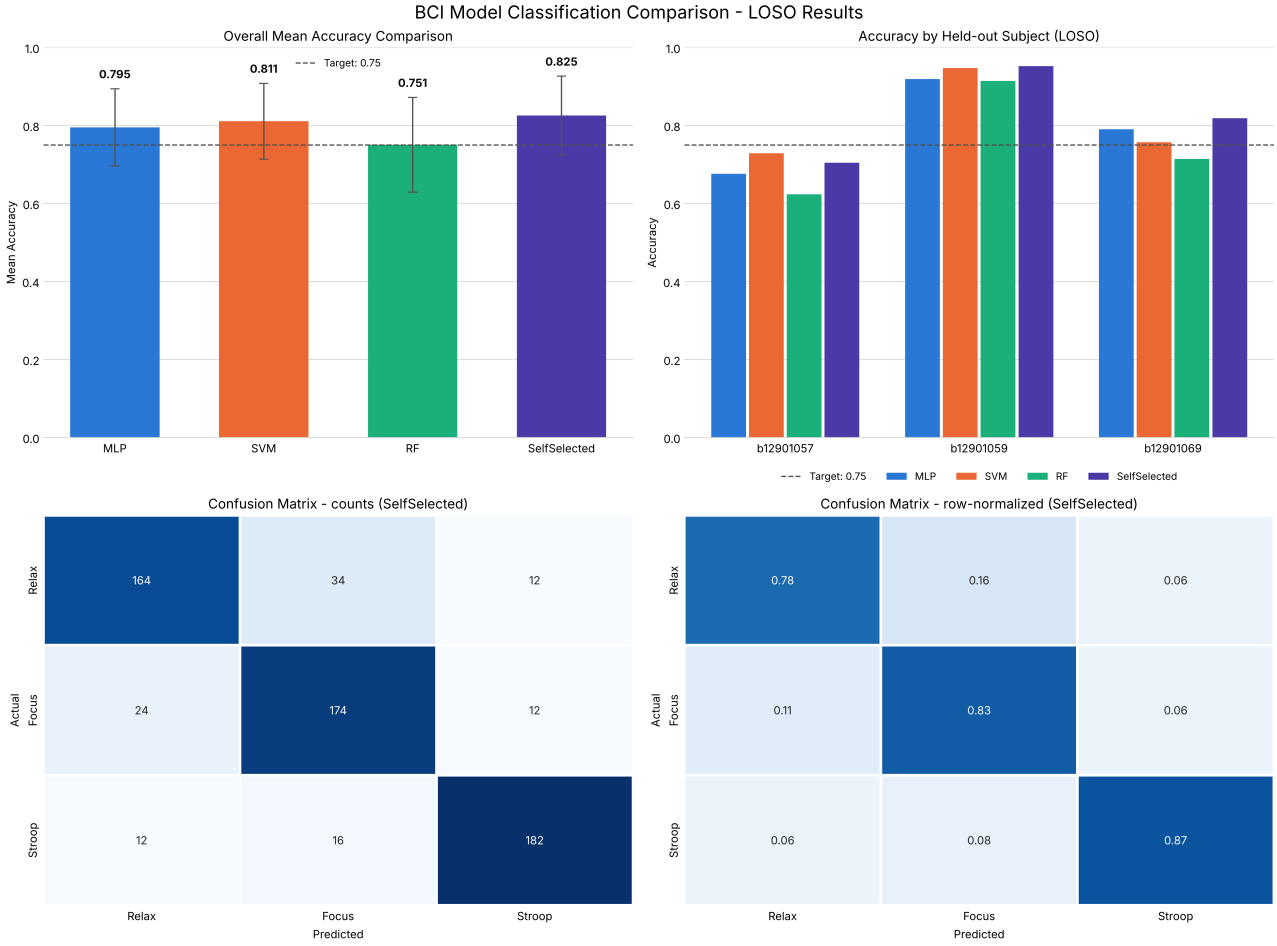


Figure 1: LOSO cross-validation results of the best model, including the accuracy of the four models, the per-subject accuracy for the 3 group members, and the overall 3×3 confusion matrix.

3.4 Result Analysis and Discussion

- **Accuracy Comparison:** Shrinkage LDA (0.825) > SVM (0.811) > MLP (0.795) > RF (0.751); all of them reached the $\geq 75\%$ target, and the best two models (LDA and SVM) also exceeded 80%. LDA suits this dataset for three reasons. After the subject-wise z-score, the class differences are mostly shifts of the mean of a few band features (Relax has the highest relative alpha, Focus the highest relative beta, Stroop the highest amplitude and theta; class means of the z-scored features over all three members, see below), which a linear boundary can separate. There are only 420 training samples for 32 features, so a model with few parameters and automatic covariance shrinkage is less prone to over-fitting than the MLP or the RF. Finally, LDA has no hyperparameter that needs tuning on only three folds.

- **Confusion Matrix Analysis:** Using the 3×3 confusion matrix in Figure 1:

  - *Easiest class: Stroop* (recall 0.867, precision 0.883). Its z-scored amplitude and kurtosis are the highest (+0.89 and +0.98 versus −0.55/−0.59 for Relax), and its relative theta and θ/β are also the highest. Stroop has the highest amplitude in each of the three members (z-scored log standard deviation +0.67 to +1.06), not only on average. This fits the expectation that reading the words and answering with the keyboard add eye and muscle activity at Fp1, on top of the frontal theta that accompanies cognitive conflict [7]. With one channel, however, we cannot tell how much of the amplitude is cortical and how much is artifact.

  - *Most confused pair: Relax and Focus.* 34 Relax segments were predicted as Focus and 24 Focus segments as Relax (58 errors), compared with 28 errors between Focus and Stroop and 24 between Relax and Stroop. Both states have a below-average amplitude (z-scored log standard deviation −0.55 and −0.34), and they differ mainly in the balance between alpha (higher in Relax, +0.95) and beta (higher in Focus), which is subtle at a single frontal electrode.

  - *Effect of our processing.* Counting the Relax and Focus errors of LDA at each ablation step (Table 4) shows where the improvement comes from. Of the 420 Relax and Focus segments, 176 were confused between the two states with the raw samples as input, 157 with the band-power features and 151 after adding the time-domain features, but only 58 after the subject-wise z-score. The two states differ mainly in the balance between alpha and beta, and this difference only becomes consistent across members once each member's baseline is removed. The time-domain features hardly help this pair (without them, the z-scored band-power features give 51 errors instead of 58), because both states have a below-average amplitude; their gain is in separating Stroop, where the errors between Relax and Stroop fall from 57 to 28. The other three models show the same trend, with 49 to 64 Relax and Focus errors in the final pipeline. No post-processing was used in the final code (Section 2), so the remaining 58 errors are still more than half of all LDA errors.

- **Inter-Subject Variability:** The model did not perform equally well on all members. With LDA, B12901059 reached 0.952, B12901069 0.819, and B12901057 only 0.705; the other models show the same ranking. For B12901057, 28 of 70 Relax segments were predicted as Focus. The class means explain part of it: B12901059 has a large alpha contrast between Relax and Focus (z-scored relative alpha +1.29 versus −0.73) and B12901069 a moderate one (+1.15 versus −0.38), while B12901057 has a weak one (+0.41 versus +0.21), and for

B12901057 the amplitude of Relax is even larger than that of Focus (−0.09 versus −0.85 in z-scored log standard deviation), the reverse of B12901059 (−1.25 versus +0.20). A model trained on the other two members therefore mislabels this member's Relax recordings. The weak alpha contrast also appears in the earlier sessions of B12901057, so it is unlikely to come from a single poor session. It more likely reflects a weak alpha response of this subject at Fp1, where alpha is much smaller than over the occipital cortex.

- **Method Effectiveness Analysis:** Based on Table 4, the *time-domain features* (+12.9 points, from 0.571 to 0.700) and the *subject-wise z-score* (+12.5 points, from 0.700 to 0.825) contributed the most, followed by *filtering* (+3.7) and the *window band-power features* (+3.4). Because the best model differs between rows, we also follow a single model: for LDA the steps give 0.338 (filtered raw samples), 0.571 (band-power features), 0.667 (time-domain features) and 0.825 (z-score), so the z-score is the largest single gain (+15.8 points) and the time-domain features add 9.6 points (Appendix B). Replacing 4096 raw samples with 24–32 physiologically motivated features is what makes a high LOSO accuracy possible, and the z-score removes the between-subject baseline differences that otherwise dominate (for instance, the median raw standard deviation of the Relax recordings is 15.8 for B12901059 but about 31 for the other two members). Note that the z-score uses the held-out member's own unlabeled data to compute the mean and standard deviation.

- **Limitations:** The data consist of a single dry electrode at Fp1 and only three subjects, so the 630 segments come from just 90 recordings and overlapping segments are correlated. Treating the 90 recordings as the independent units, a binomial approximation gives a 95% confidence interval of roughly ±8 percentage points for the best accuracy, and each fold tests a single person only. The segment length, the features and the normalization were selected using the same three subjects, so the reported accuracy is optimistic. The subject-wise z-score also needs a recording of the target user before the model can be used, so it does not give a calibration-free subject-independent model. A more reliable conclusion would need more subjects (ideally more than ten), more electrode sites, a recording protocol that controls eye movement and key presses, and a validation on subjects never used for choosing the features or hyperparameters.

4 Discussion

Recording with the BrainLink headset. Each member recorded 30 blocks of 20 s (ten rounds of Relax, Focus and Stroop). Pairing the headset and selecting the serial port caused no problems. However, in one session we only noticed after all rounds had been recorded that the headset had not actually been connected, so the whole session had to be repeated. A few files also turned out to be slightly shorter than 20 s, so we recorded those rounds again. Since the first session had likely been disturbed by the crowded, noisy lab and a tired subject, we compared the spectra with the reference examples before keeping a session. The dry electrode at Fp1 is sensitive to contact quality and movement, and both the Focus and the Stroop tasks are answered with key presses. In our data, the Stroop recordings show the largest amplitude, and also the highest 90th percentile and maximum of the 1 s window amplitude, which is partly useful for classification but cannot be separated from muscle and eye artifacts with a single channel. In the first rounds of one member's session, we observed particularly large artifacts (the standard deviation of five early recordings of B12901069 was 1.7 to 2.8 times the median

over all files), so the early rounds of a session seem to be less reliable. We kept all recordings, used the filtering described in Section 2 and relied on the distribution statistics of the window features (mean, standard deviation, 90th percentile and maximum).

Software challenges. The first version of our pipeline used one Welch spectrum per segment with 21 features (relative and log band power, ratios and time-domain statistics). It reached only about 60% LOSO accuracy at best, because the absolute and relative band powers differ strongly between members, so a model trained on two members does not transfer to the third. We addressed this with two changes: (i) the distribution of 1 s window features inside each segment, which keeps short bursts that a single spectrum averages away, and (ii) a subject-wise z-score of the features. The framework calls `extract_features` on segments without a subject ID, so the normalization had to be implemented inside Part 2 by building a segment-to-feature table once and normalizing each subject with its own statistics. Features that looked helpful in isolation, such as relative band power alone or additional spectral ratios, did not help across subjects until the normalization was added.

Limitations and future work. A single dry electrode at Fp1 captures frontal activity only and mixes EEG with eye-blink and muscle signals. Alpha reactivity, which is the main cue for Relax, is much weaker at Fp1 than over the occipital cortex, and B12901057 shows only a weak alpha difference between Relax and Focus. With three subjects the results are also sensitive to a single member. Future work could (i) record more subjects and more electrode positions, (ii) control or label the eye and key-press artifacts (for example with a blink-removal step or a hands-free response), (iii) test normalization that only needs a short resting baseline from the new user, or domain-adaptation methods, to reduce the need for a recording of the target user, and (iv) evaluate on subjects not used for any design decision.

5 Conclusion

In this experiment we recorded our own single-channel Fp1 EEG with a BrainLink headset (three members, 30 recordings each) and built a pipeline to classify the Relax, Focus and Stroop states under Leave-One-Subject-Out cross-validation. Each 8 s segment (75% overlap) was denoised with DC removal, a 60 Hz notch and a 1–45 Hz band-pass, and described by 32 features: the distribution of 1 s window features (relative theta/alpha/beta power, θ/β and α/β ratios, amplitude and kurtosis), followed by a subject-wise z-score. Four classifiers were compared on exactly the same features. Shrinkage LDA performed best with a mean LOSO accuracy of 82.5%, followed by the SVM (81.1%), the MLP (79.5%) and the Random Forest (75.1%), whose hyperparameters were selected by parameter experiments. The ablation study shows that the time-domain features and the subject-wise normalization contributed the most (about +13 points each for the best model of each step; the normalization is the largest single gain, +15.8 points, for LDA alone), followed by filtering and the window band-power features. The goal of $\geq 75\%$ accuracy was met by all four classifiers, and the two best models (LDA and SVM) also exceeded 80%. The accuracy differs strongly between members (0.705 to 0.952 for the best model), and the results are based on only three subjects, so they should be regarded as optimistic estimates rather than a general subject-independent performance.

6 Team Roles and Responsibilities

Table 5: Team roles and responsibilities

Team Member	Primary Role	Key Responsibilities
楊程亘 (B12901057)	Data Processing	• Recorded own 30 BrainLink recordings (Part 1). • Data preprocessing and segmentation. • Model optimization. • Report writing.
陳裕穎 (B12901059)	Model Training	• Recorded own 30 BrainLink recordings (Part 1). • Model training and LOSO evaluation of the four classifiers, hyperparameter tuning. • Choice of the self-selected model and comparison with other candidate classifiers. • Model optimization. • Report writing.
段奕鳴 (B12901069)	Data Analysis	• Recorded own 30 BrainLink recordings (Part 1). • Data quality and analysis of the recordings. • Model optimization. • Report writing.

Appendix A Additional Information

The source code and the dataset are available on GitHub: github.com/VK-shu/BCI. The repository contains the submitted `main.py`, the recordings in `bci_dataset_115-1/`, `tools/check_data.py` (checks the file names and the length of every recording), `exp_testing/ablation.py` (reproduces Table 4 and Table 7), the parameter experiments in `exp_testing/` (Appendix B) and the LaTeX source of this report.

Appendix B Experimental Results

B.1 Reproducibility

All numbers in this appendix are produced by scripts in the repository, using the final features of `main.py` (which is only imported, never modified or run by them) and Leave-One-Subject-Out validation with the `StandardScaler` fitted on the training members only, exactly as in `MultiModelBCIClassifier`. All random elements use `Config.RANDOM_STATE` = 42 unless a seed sweep is stated. Every script was run twice from scratch and the two result files were identical line by line.

- `exp_testing/ablation.py` → `results_ablation.txt`: Tables 4 and 7.

- `exp_mlp.py`, `exp_svm.py`, `exp_rf.py` and `exp_lda.py` in `exp_testing/` → the matching `results_*.txt` files: Tables 8–14 (classifier experiments).
- `exp_testing/exp_pipeline.py` → `results_pipeline.txt`: Tables 15 and 16.
- Run with `uv run python <script>` from the project root. This uses the course environment defined by `pyproject.toml` and `uv.lock` (Python 3.11, NumPy 2.4.3, scikit-learn 1.8.0, SciPy 1.17.1); other library versions may change the last digit of some values.

In the tables, **bold** marks the setting used in `main.py` and † the framework's initial value (`main_original.py`). Each cell is the mean LOSO accuracy over the three held-out members, and one parameter is varied at a time while the others stay at their final values, unless the row says otherwise.

B.2 Per-subject and Ablation Results

Table 6: Per-subject LOSO accuracy of the four models (final pipeline)

Held-out subject	**MLP**	**SVM**	**RF**	**Shrinkage LDA**
B12901057	0.676	0.729	0.624	0.705
B12901059	0.919	0.948	0.914	0.952
B12901069	0.790	0.757	0.714	0.819
Mean ± std	0.795 ± 0.099	0.811 ± 0.097	0.751 ± 0.121	0.825 ± 0.101

Table 7: Ablation study: mean LOSO accuracy ± std of every model for each configuration (`ablation.py`; each row adds one step to the row above, the subject-wise z-score is off until row 5)

Configuration	**#feat.**	**MLP**	**SVM**	**RF**	**LDA**
Raw samples	4096	0.305 ± 0.025	0.500 ± 0.014	0.348 ± 0.025	0.303 ± 0.018
+ Filtering (raw samples)	4096	0.308 ± 0.020	0.537 ± 0.036	0.376 ± 0.022	0.338 ± 0.004
+ Window band power	24	0.533 ± 0.068	0.425 ± 0.077	0.508 ± 0.075	0.571 ± 0.068
+ Time-domain features	32	0.700 ± 0.056	0.535 ± 0.057	0.641 ± 0.076	0.667 ± 0.037
+ Subject-wise z-score (final)	32	0.795 ± 0.099	0.811 ± 0.097	0.751 ± 0.121	0.825 ± 0.101
Extra: band power + z-score	24	0.763 ± 0.150	0.779 ± 0.140	0.765 ± 0.165	0.803 ± 0.121

B.3 Classifier Experiments

Table 8: MLP: sensitivity of the LOSO accuracy to each hyperparameter (`exp_mlp.py`)

Model	Parameter	Tested values: mean accuracy
MLP	hidden layers	(16,): 0.794; (32,): 0.741; (64,): 0.762; (128,): 0.781; (256,): 0.787; (32, 16): 0.757; (64, 32): 0.770; (128, 64): 0.795; (128, 64, 32): 0.784†; **(256, 128, 64): 0.795**; (64, 64, 64): 0.778; (256, 128, 64, 32): 0.798
	learning rate	0.0001: 0.784; **0.0005: 0.795**; 0.001: 0.789; 0.002: 0.775†; 0.0025: 0.779; 0.005: 0.798; 0.01: 0.778; 0.02: 0.730; 0.05: 0.759
	alpha (L2)	0.0: 0.797; 1e-05: 0.798; 0.0001: 0.797; 0.001: 0.800; 0.005: 0.800†; 0.01: 0.800; **0.05: 0.795**; 0.1: 0.795; 0.5: 0.794; 1.0: 0.792; 5.0: 0.778
	max iterations	10: 0.795; 25: 0.795; 50: 0.795; 100: 0.795; **150: 0.795**†; 300: 0.795; 500: 0.795; 1000: 0.795
	activation*	**relu: 0.795**; tanh: 0.806; logistic: 0.768; identity: 0.779
	batch size*	8: 0.779; 16: 0.797; **32: 0.795**; 64: 0.797; 128: 0.767; 256: 0.768
	solver*	**adam: 0.795**; sgd: 0.765; lbfgs: 0.771
	early stopping*	**True: 0.795**; False: 0.789
	random seed 0–9	0: 0.800; 1: 0.790; 2: 0.808; 3: 0.783; 4: 0.798; 5: 0.794; 6: 0.794; 7: 0.779; 8: 0.781; 9: 0.800; **mean 0.793 ± 0.009** (the setting in `main.py` uses seed 42: 0.795)

* not a `Config` setting; varied here for completion.

Table 9: MLP: ten best settings of the joint grid hidden layers × learning rate × alpha (125 settings, mean over seeds 0, 1, 2, `max_iter` = 150), and the setting used in `main.py`

Rank	Hidden layers / learning rate / alpha	Mean accuracy
1	(256, 128, 64) / lr=0.0005 / alpha=0.05	0.799
2	(256, 128, 64) / lr=0.0025 / alpha=0.005	0.798
3	(128, 64) / lr=0.0005 / alpha=0.5	0.797
4	(128, 64) / lr=0.001 / alpha=0.5	0.797
5	(256, 128, 64) / lr=0.0005 / alpha=0.005	0.796
6	(256, 128, 64) / lr=0.0005 / alpha=0.5	0.796
7	(256, 128, 64) / lr=0.001 / alpha=2.0	0.796
8	(128, 64) / lr=0.0005 / alpha=0.0001	0.795
9	(128, 64) / lr=0.0005 / alpha=0.005	0.795
10	(128, 64) / lr=0.001 / alpha=0.0001	0.795
—	**(256, 128, 64) / lr=0.0005 / alpha=0.05** (`main.py`)	0.799

Table 10: SVM: sensitivity of the LOSO accuracy to each hyperparameter (`exp_svm.py`)

Model	Parameter	Tested values: mean accuracy
SVM	kernel	linear: 0.800; **rbf: 0.811**†; poly: 0.510; sigmoid: 0.783
	C (rbf kernel)	0.001: 0.811; **0.01: 0.811**; 0.05: 0.811; 0.1: 0.803; 0.2: 0.803; 0.3: 0.800; 0.5: 0.797; 0.75: 0.789; 1.0: 0.781†; 2.0: 0.762; 5.0: 0.751; 10.0: 0.738; 50.0: 0.748; 100.0: 0.748; 1000.0: 0.748
	gamma (rbf kernel)	**scale: 0.811**; auto: 0.811; 0.0003: 0.771; 0.001: 0.773; 0.003: 0.792; 0.01: 0.794; 0.03: 0.813; 0.1: 0.751; 0.3: 0.468; 1.0: 0.335
	C (linear kernel)	0.0001: 0.770; 0.001: 0.808; 0.01: 0.800; 0.1: 0.779; 1.0: 0.767; 10.0: 0.748
	degree (poly kernel)	2: 0.467; 3: 0.510; 4: 0.349; 5: 0.386
	class weight	**None: 0.811**; balanced: 0.811

Table 11: SVM: joint grid kernel × C (mean LOSO accuracy)

Kernel	0.005	0.01	0.02	0.05	0.1	0.2	0.3	0.5	1.0	2.0	5.0	10.0
rbf	0.811	**0.811**	0.811	0.811	0.803	0.803	0.800	0.797	0.781†	0.762	0.751	0.738
linear	0.790	0.800	0.794	0.790	0.779	0.768	0.768	0.762	0.767	0.767	0.759	0.748
sigmoid	0.783	0.783	0.794	0.808	0.808	0.795	0.806	0.798	0.776	0.719	0.702	0.697
poly	0.400	0.510	0.506	0.635	0.698	0.729	0.735	0.749	0.760	0.756	0.756	0.746

Table 12: Random Forest: sensitivity of the LOSO accuracy to each hyperparameter (`exp_rf.py`)

Model	Parameter	Tested values: mean accuracy
RF	n_estimators	5: 0.673; 10: 0.702; 25: 0.741; 50: 0.729; 100: 0.748†; 200: 0.748; 300: 0.759; **500: 0.751**; 1000: 0.752
	max_depth	1: 0.690; 2: 0.754; 3: 0.763; 4: 0.759; **5: 0.751**; 6: 0.748; 8: 0.751; 10: 0.756; 15: 0.757; 20: 0.757; None: 0.757†
	min_samples_leaf*	**1: 0.751**; 2: 0.748; 3: 0.751; 5: 0.751; 10: 0.762; 20: 0.762; 40: 0.749
	min_samples_split*	**2: 0.751**; 5: 0.752; 10: 0.746; 20: 0.737; 50: 0.741
	max_features*	**sqrt: 0.751**; log2: 0.751; 0.1: 0.773; 0.25: 0.735; 0.5: 0.711; 0.75: 0.697; None: 0.687
	criterion*	**gini: 0.751**; entropy: 0.757; log_loss: 0.757
	bootstrap / max_samples*	**bootstrap=True: 0.751**; bootstrap=False: 0.754; max_samples=0.3: 0.756; max_samples=0.5: 0.757; max_samples=0.8: 0.746
	class_weight*	**None: 0.751**; balanced: 0.751; balanced_subsample: 0.751
	random seed 0–9	0: 0.763; 1: 0.749; 2: 0.767; 3: 0.762; 4: 0.760; 5: 0.744; 6: 0.754; 7: 0.762; 8: 0.759; 9: 0.759; **mean 0.758 ± 0.006** (seed 42 in `main.py`: 0.751)

* not a `Config` setting; varied here for completion.

Table 13: Random Forest: joint grid n_estimators × max_depth (mean LOSO accuracy over seeds 0–4)

n_estimators	**3**	**4**	**5**	**6**	**8**	**10**	**None**
50	0.750	0.749	0.753	0.740	0.741	0.739	0.737
100	0.756	0.755	0.757	0.747	0.743	0.742	0.744†
300	0.757	0.755	0.751	0.752	0.745	0.747	0.745
500	0.757	0.758	**0.760**	0.754	0.750	0.748	0.749

Table 14: Self-selected model (shrinkage LDA): settings and alternative candidate classifiers (`exp_lda.py`)

Model	**Parameter**	**Tested values: mean accuracy**
LDA	solver, shrinkage	svd: 0.813; lsqr, none: 0.813; **lsqr, auto: 0.825**; eigen, none: 0.813; eigen, auto: 0.825
	manual shrinkage	0.0: 0.813; 0.01: 0.816; 0.05: 0.827; 0.1: 0.822; 0.2: 0.817; 0.3: 0.816; 0.4: 0.816; 0.5: 0.814; 0.6: 0.811; 0.7: 0.803; 0.8: 0.795; 0.9: 0.790; 1.0: 0.787
	class priors	priors=None (from data): 0.825; priors=uniform (1/3 each): 0.825
QDA	reg_param	0.0: 0.622; 0.1: 0.746; 0.3: 0.765; 0.5: 0.786; 0.7: 0.784; 0.9: 0.786; 1.0: 0.787
Logistic regr.	C	0.001: 0.787; 0.01: 0.800; 0.1: 0.805; 1.0: 0.802; 10.0: 0.779; 100.0: 0.746
k-NN	n_neighbors	1: 0.735; 3: 0.759; 5: 0.775; 9: 0.800; 15: 0.806; 25: 0.806; 41: 0.798; 61: 0.798; 15 (distance-weighted): 0.808
Others		GaussianNB: 0.757; RidgeClassifier alpha=1.0: 0.824; RidgeClassifier alpha=100.0: 0.825; RidgeClassifier alpha=1000.0: 0.816; LinearSVC C=0.001: 0.817; LinearSVC C=0.01: 0.819; LinearSVC C=0.1: 0.819; ExtraTrees (300 trees): 0.783; GradientBoosting (100, depth 3): 0.730

B.4 Pipeline Experiments

Table 15: Effect of the segment length and the overlap ratio on the LOSO accuracy (`exp_pipeline.py`)

Length (s)	Overlap	Samples/subject	MLP	SVM	RF	LDA
3	0.50	360	0.689	0.742	0.715	0.719
5	0.50	210	0.751	0.783	0.763	0.795
5	0.75	390	0.756	0.782	0.758	0.782
8	0.50	120	0.781	0.811	0.744	0.806
8	**0.75**	**210**	**0.795**	**0.811**	**0.751**	**0.825**
10	0.50	90	0.781	0.804	0.774	0.852
10	0.75	150	0.807	0.807	0.749	0.836

Table 16: Feature selection (`SelectKBest`, k of the 32 features kept; $k = 32$ is the final setting without selection)

k	MLP	SVM	RF	LDA
8	0.762	0.765	0.721	0.783
12	0.762	0.778	0.714	0.786
15	0.771	0.789	0.733	0.810
20	0.763	0.798	0.738	0.811
25	0.783	0.806	0.759	0.806
32	**0.795**	**0.811**	**0.751**	**0.825**

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